Sahaj Bhadja

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Summary

Front-end engineer with 2+ years building **React/TypeScript** apps, **WebGL/three.js/GLSL** visualizations, and performance-focused autonomy tooling.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Built Azure-backed AI education platform with normalized schemas, row-level isolation, and indexed dashboard queries, cutting latency from 1.2 seconds to 350 milliseconds for 500+ campus users daily.
- Reworked asynchronous LLM orchestration with FastAPI workers, job-state transitions, and failure isolation, tripling throughput for 1K+ AI jobs under concurrent instructor-facing web workloads during production release cycles.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built React/Flask grading workflows for uploads, rubrics, grading history, and model selection, supporting 150+ instructors and 10K+ daily requests through production REST APIs and instructor dashboards reliably.
- Designed concurrent PDF, DOCX, and ZIP ingestion with ThreadPoolExecutor, OCR, and multimodal routing, keeping grading responsive while reducing feedback turnaround 60% during peak load for instructors daily.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built a Decision Transformer for autonomous-driving offline RL, improving planning 18% in simulation while processing 100K+ trajectory logs through reproducible Python evaluation pipelines for autonomy research experiments.
- Built a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% across simulated driving baselines for planning research workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built React/Node resource-management workflows with WebSockets, REST APIs, and modular services for real-time records, synchronizing 2M+ cross-location entries with 33% fewer conflicts during staged production rollouts safely.
- Implemented responsive CRM modules for authentication, resources, billing, leads, permissions, and notifications, using Jenkins CI/CD and Docker staging to cut deployment time 50% across release cycles safely.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built React/TypeScript booking flows for time-slot selection, reservation tracking, coupon pricing, and payment-status updates across 50K+ service records with sub-200-millisecond APIs for client demos and reviews successfully.
- Designed Figma-led architecture for 12+ integrations, aligning Flask APIs, Firebase synchronization, webhooks, and cross-platform delivery before implementation to reduce design errors 25% for stakeholder reviews early successfully.

Skills

Graphics & Web: JavaScript, TypeScript, React, Next.js, HTML, CSS, WebGL, OpenGL, Three.js, GLSL, Shaders, 3D Math
Visualization: Linear Algebra, Scene Graphs, Camera Frustums, Trajectories, Browser Performance, Responsive UI, Figma, Debugging
Backend & Data: Python, C++, SQL, Node.js, Flask, FastAPI, REST APIs, WebSockets, MongoDB, Firebase
AI & Delivery: PyTorch, Transformers, Offline RL, HD Maps, Autonomous Simulation, Git, CI/CD, Docker, Unit Testing, Agile/Scrum

Projects

DriveScene Renderer | *JavaScript, three.js, WebGL, GLSL, React, 3D Math*

- Created a three.js autonomous-scene renderer with WebGL, GLSL shaders, and React controls for lidar-style points, camera frustums, trajectories, and debugging overlays inside browser simulation views locally interactive.

PiXMod Live | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*

- Built a responsive Next.js image platform with CLIP search, SAM segmentation, Stable Diffusion editing, MongoDB storage, and Stripe billing, processing 1K+ edited visual assets across workflows online.

CPU-scheduling Simulator Live | *C++, React, JavaScript, EJ-2*

- Implemented React controls and EJ-2 Gantt visualization for FCFS, SJF, SRT, RR, priority, and MLQ scheduling, calculating turnaround metrics across 1K+ process combinations with dynamic UI updates.